PONMATHISH T

DAY - 02

10 MARKS

1. What is a Computer Network?

A **computer network** is a group of two or more computers and other devices connected together to **share data, information, resources, and services**.

Computers in a network can communicate with each other using wired or wireless communication methods.

Examples: Internet, Local Area Network (LAN), Wi-Fi network.

Main advantages:

- Sharing files and data
- Sharing hardware such as printers
- Communication between users
- Sharing Internet connections
- Centralized management of resources

2. What is the difference between a Client and a Server?

A **client** and a **server** are two important components of a computer network. They communicate with each other to provide and access services and resources.

Client

A **client** is a computer or device that **requests services, data, or resources** from another computer called a server. Clients are usually used directly by users.

Examples: Desktop computer, laptop, smartphone, web browser.

Server

A **server** is a computer or system that **provides services, data, or resources** to clients over a network. It usually manages and responds to requests from multiple clients.

Examples: Web server, file server, email server, database server.

Difference between Client and Server

Client	**Server**
1. Requests services or resources.	1. Provides services or resources.
2. Usually used by an end user.	2. Usually manages network services.
3. Sends requests to the server.	3. Receives and processes requests.
4. Receives responses from the server.	4. Sends responses to clients.
5. Usually has fewer resources.	5. Usually has greater processing and storage capacity.
6. Can connect to one or more servers.	6. Can serve many clients simultaneously.
7. Example: Web browser.	7. Example: Web server.
8. It depends on servers for certain services.	8. It provides services to clients.
9. Client software runs on the user's device.	9. Server software runs on the server system.
10. Example: A user's computer requesting a webpage.	10. Example: A server sending the requested webpage.

Example

When a user opens a website using a **web browser**, the browser acts as the **client**. It sends a request to the **web server**. The web server processes the request and sends the webpage back to the client.

In simple terms:

Client → Requests the service

Server → Provides the service

3. What is a LAN?

LAN stands for **Local Area Network**. It is a computer network that connects computers and other devices within a **small geographical area**, such as a home, office, school, college, or building.

A LAN allows connected devices to **share data, files, hardware, software, and Internet connections** with each other.

Features of LAN

1. **Small geographical area:** It covers a limited area such as a room, building, or campus.
2. **High speed:** LAN provides high-speed communication between connected devices.
3. **Resource sharing:** Users can share printers, files, storage, and other resources.
4. **Data sharing:** Data can be transferred easily between computers.
5. **Low cost:** LAN is relatively inexpensive to install and maintain.
6. **Private network:** It is generally owned and controlled by an organization or individual.
7. **Easy management:** Network administrators can manage connected devices centrally.
8. **Wired or wireless:** LAN can use Ethernet cables or wireless technologies such as Wi-Fi.
9. **Reliable communication:** Devices can communicate efficiently within the network.

10. **Internet sharing:** Multiple computers can access the Internet through a common connection.

Examples

- A **school computer laboratory** where all computers are connected.
- An **office network** connecting computers and printers.
- A **home Wi-Fi network** connecting phones, laptops, and smart devices.

Advantages of LAN

- Fast data transfer
- Easy file and resource sharing
- Sharing of printers and other hardware
- Internet connection sharing
- Easy communication between users

Conclusion

A **LAN is a network used to connect computers and devices within a limited geographical area**. It provides fast communication and enables users to share data and resources efficiently.

4. What is a WAN?

WAN stands for **Wide Area Network**. It is a computer network that connects computers, networks, and other devices over a **large geographical area**, such as different cities, states, countries, or even continents.

A WAN is used to connect multiple **LANs (Local Area Networks)** together so that users in different locations can communicate and share information.

Features of WAN

1. **Large geographical coverage:** WAN can cover cities, countries, or continents.

2. **Connects multiple LANs:** It connects networks located in different places.
3. **Long-distance communication:** It allows communication between distant locations.
4. **Uses communication links:** It may use fiber-optic cables, satellites, leased lines, and wireless connections.
5. **Lower speed than LAN:** Generally, WAN communication may be slower than a local LAN.
6. **Higher cost:** Setting up and maintaining a WAN is usually more expensive than a LAN.
7. **Centralized services:** Organizations can share applications, databases, and resources between branches.
8. **Requires network devices:** Routers, switches, and other networking equipment are commonly used.
9. **Can be public or private:** WANs may be operated by organizations or telecommunications providers.
10. **Supports remote communication:** Users can communicate and access resources from different locations.

Examples of WAN

- **The Internet** is the largest example of a WAN.
- A bank connecting its branches in different cities.
- A company connecting offices in different countries.

Advantages of WAN

- Connects geographically distant offices.
- Enables long-distance communication.
- Allows sharing of data and resources.

- Supports centralized databases and applications.
- Helps organizations communicate between branches.

LAN vs WAN

LAN	**WAN**
Covers a small area.	Covers a large geographical area.
Usually has higher speed.	Generally has lower speed than LAN.
Less expensive to set up.	More expensive to set up.
Example: School computer lab.	Example: Internet.

Conclusion

A **WAN is a network that connects computers and smaller networks over a large geographical area**. It is widely used by organizations, businesses, and telecommunication providers to enable communication and resource sharing between distant locations.

5. What is the Purpose of a Switch?

A **switch** is a networking device used to **connect multiple computers and other devices within a network**, especially in a LAN (Local Area Network). Its main purpose is to **receive data and forward it to the correct destination device**.

Functions and Purposes of a Switch

1. **Connects network devices:** A switch connects computers, printers, servers, and other devices in a LAN.
2. **Forwards data:** It receives data packets and sends them to the intended device.
3. **Uses MAC addresses:** A switch uses the **MAC address** of devices to determine where data should be sent.

4. **Reduces unnecessary traffic:** It sends data only to the required destination instead of sending it to every device.
5. **Improves network performance:** By reducing unnecessary traffic, it helps improve network speed and efficiency.
6. **Enables communication:** It allows devices connected to the same network to communicate with each other.
7. **Supports multiple devices:** A switch can provide multiple ports for connecting several devices.
8. **Creates a LAN:** Switches are commonly used as the central connection point in wired LANs.
9. **Provides efficient data transfer:** It can handle communication between multiple devices at the same time.
10. **Improves network security:** Managed switches can provide features such as VLANs and access control to help control network access.

Example

In a **computer laboratory**, 20 computers can be connected to a switch using Ethernet cables. When one computer sends data to another, the switch identifies the destination device and forwards the data to the appropriate port.

Conclusion

A **switch is an important networking device that connects devices in a LAN and efficiently forwards data to the correct destination using MAC addresses**. It helps improve network performance, communication, and resource sharing.

6. What is the Purpose of a Router?

A **router** is a networking device that **connects two or more different networks** and forwards data packets between them. It determines the best path for data to travel from the source network to the destination network.

Functions and Purposes of a Router

1. **Connects different networks:** A router connects separate networks, such as a home LAN to the Internet.
2. **Forwards data packets:** It receives data packets and sends them toward their destination.
3. **Uses IP addresses:** A router uses **IP addresses** to identify the source and destination networks.
4. **Selects the best path:** It determines an appropriate route for data to reach its destination.
5. **Connects LAN to WAN:** Routers can connect a local network to a wider network such as the Internet.
6. **Provides Internet access:** A home router allows multiple devices to share a single Internet connection.
7. **Separates networks:** It keeps different networks logically separated and controls communication between them.
8. **Provides security:** Many routers include firewall and access-control features to protect the network.
9. **Supports wireless communication:** Wireless routers can provide Wi-Fi connectivity to computers, phones, and other devices.
10. **Manages network traffic:** A router helps direct traffic efficiently between different networks.

Example

In a home network, computers and smartphones connect to a **router** through Wi-Fi or Ethernet. The router connects the home LAN to the **Internet** and forwards requests from the devices to the appropriate Internet destinations.

Switch vs Router

Switch	Router
Connects devices within a LAN.	Connects different networks.
Mainly uses MAC addresses.	Mainly uses IP addresses.
Forwards frames within a network.	Routes packets between networks.
Example: Connecting computers in an office.	Example: Connecting an office network to the Internet.

Conclusion

A **router is an essential networking device that connects different networks and forwards data packets using IP addresses**. It helps devices communicate with other networks and provides access to services such as the Internet.

7. What is a Firewall?

A **firewall** is a **network security system** that monitors and controls incoming and outgoing network traffic based on predefined security rules. Its main purpose is to **protect computers and networks from unauthorized access, malicious traffic, and cyber threats**.

Functions and Purposes of a Firewall

1. **Controls network traffic:** A firewall monitors data entering and leaving a network.
2. **Prevents unauthorized access:** It blocks unauthorized users from accessing protected systems.
3. **Filters data packets:** It examines network packets and allows or blocks them according to security rules.
4. **Protects against cyber attacks:** It helps protect networks from threats such as malicious connections and unauthorized access attempts.

5. **Blocks suspicious traffic:** Unwanted or suspicious network traffic can be denied.
6. **Controls access to services:** A firewall can allow or restrict access to specific applications, ports, or services.
7. **Protects private networks:** It acts as a security barrier between a trusted internal network and an untrusted network such as the Internet.
8. **Monitors network activity:** It can record and log network connections and security events.
9. **Supports network security policies:** Administrators can create rules to control which traffic is permitted.
10. **Reduces security risks:** By filtering unwanted traffic, a firewall helps reduce the risk of unauthorized access and attacks.

Types of Firewalls

- **Hardware Firewall:** A physical device that protects a network.
- **Software Firewall:** Software installed on a computer or server.
- **Network Firewall:** Protects an entire network from unwanted traffic.
- **Host-based Firewall:** Protects an individual computer or device.

Example

When a computer connected to the Internet receives an unwanted connection request, the **firewall checks the request against its security rules**. If the traffic is not permitted, the firewall blocks it.

Conclusion

A **firewall acts as a security barrier between a trusted network and an untrusted network**. It monitors and filters network traffic and helps prevent unauthorized access and cyber threats.

8. What is an IP Address?

An **IP address (Internet Protocol address)** is a **unique logical address assigned to a device connected to a computer network**. It is used to identify the device and enable communication between devices over a network.

Functions and Purposes of an IP Address

1. **Identifies a device:** An IP address identifies a computer, smartphone, server, printer, or other network device.
2. **Enables communication:** It allows devices to communicate with each other over a network.
3. **Identifies the destination:** Routers use IP addresses to determine where data packets should be delivered.
4. **Provides network information:** An IP address indicates the network and device associated with a connection.
5. **Supports Internet communication:** Devices use IP addresses to communicate across the Internet.
6. **Used by routers:** Routers examine destination IP addresses when forwarding packets.
7. **Can be assigned automatically:** A device can receive an IP address automatically using **DHCP**.
8. **Can be static or dynamic:** IP addresses may be permanently assigned or temporarily assigned.
9. **Used in different versions:** The two main versions are **IPv4** and **IPv6**.
10. **Helps deliver data correctly:** IP addressing ensures that data packets can be routed toward the correct destination.

Types of IP Addresses

1. IPv4

IPv4 uses a **32-bit address** and is normally written in four decimal numbers separated by dots.

Example: 192.168.1.10

2. IPv6

IPv6 uses a **128-bit address** and is written using hexadecimal numbers.

Example: 2001:db8::1

Public and Private IP Addresses

- **Public IP:** Used to identify a network or device on the Internet.
- **Private IP:** Used within a local network, such as a home or office network.

Example

When you access a website, your device sends a request containing network addressing information. Routers use the destination IP address to forward the data toward the correct server.

Conclusion

An **IP address is a logical address used to identify devices and enable data communication over computer networks**. It is an essential part of networking and allows routers to deliver data to the correct destination.

9. What is a MAC Address?

A **MAC address (Media Access Control address)** is a **unique hardware address assigned to a network interface of a device**, such as a computer, smartphone, or network printer. It is mainly used to identify devices within a **Local Area Network (LAN)**.

A MAC address operates at the **Data Link Layer (Layer 2)** of the OSI model.

Features and Functions of a MAC Address

1. **Unique identification:** A MAC address identifies a specific network interface on a device.
2. **Hardware address:** It is associated with the network interface hardware, such as an Ethernet or Wi-Fi adapter.
3. **Used in LANs:** MAC addresses are mainly used for communication within a local network.
4. **Works at Layer 2:** MAC addressing operates at the **Data Link Layer** of the OSI model.
5. **Used by switches:** Network switches use MAC addresses to forward data frames to the correct device.
6. **Usually 48 bits:** A traditional MAC address is **48 bits (6 bytes)** long.
7. **Written in hexadecimal:** It is normally represented using hexadecimal numbers.
8. **Example format:** 00:1A:2B:3C:4D:5E
9. **Different from an IP address:** A MAC address identifies a network interface, while an IP address is a logical network address.
10. **Used for local communication:** Devices use MAC addresses when exchanging Ethernet or Wi-Fi frames within a local network.

Structure of a MAC Address

A traditional 48-bit MAC address contains **six groups of two hexadecimal digits**.

Example:

00:1A:2B:3C:4D:5E

The first part generally identifies the manufacturer, while the remaining part identifies the network interface.

MAC Address vs IP Address

MAC Address	IP Address
Hardware/link-layer address	Logical/network-layer address
Mainly used within a local network	Used for communication across networks
Operates at OSI Layer 2	Operates at OSI Layer 3
Used by switches	Used by routers
Example: 00:1A:2B:3C:4D:5E	Example: 192.168.1.10

Conclusion

A **MAC address is a unique hardware-level address used to identify a network interface and enable communication within a local network**. Switches use MAC addresses to deliver network frames to the correct device.

10. What is a Default Gateway?

A **default gateway** is a **network device, usually a router, that acts as an access point between a local network and other networks**. When a computer needs to communicate with a device outside its own local network, it sends the data to the default gateway.

Functions and Purposes of a Default Gateway

1. **Connects different networks:** It connects a local network to other networks.
2. **Provides access to the Internet:** In most home and office networks, the router acts as the default gateway for Internet access.
3. **Forwards packets:** It receives packets destined for another network and forwards them to the appropriate network.
4. **Acts as an exit point:** It serves as the exit point for devices communicating outside their local network.

5. **Works with IP addresses:** The gateway uses IP addressing to determine where packets should be forwarded.
6. **Used when destination is outside the LAN:** If the destination device is not on the local network, the packet is sent to the default gateway.
7. **Usually a router:** A router commonly performs the role of a default gateway.
8. **Configured on network devices:** Computers and other devices normally have a default gateway configured as part of their network settings.
9. **Enables communication between networks:** It allows devices on one network to communicate with devices on other networks.
10. **Essential for external communication:** Without a suitable default gateway, a device may communicate within its local network but cannot normally reach other networks.

Example

Suppose a computer has the IP address:

IP Address: 192.168.1.10
Default Gateway: 192.168.1.1

If the computer wants to communicate with another device on the Internet, it sends the data to **192.168.1.1**. The router then forwards the data toward its destination.

Default Gateway vs Router

A **router** is a networking device that connects different networks and routes packets. A **default gateway** is the gateway address used by a device when it needs to send traffic to a destination outside its local network. In many networks, the router's local IP address serves as the default gateway.

Conclusion

A **default gateway is the device or address through which a computer sends data to networks outside its own local network**. It is essential for communication between a local network and external networks such as the Internet.

11. What is DNS?

DNS stands for **Domain Name System**. It is a system used to **translate human-readable domain names into IP addresses** so that computers can locate and communicate with the correct servers on a network or the Internet.

For example, when a user enters **www.google.com**, DNS helps find the IP address associated with that domain name.

Functions and Purposes of DNS

1. **Translates domain names:** DNS converts domain names into IP addresses.
2. **Makes websites easy to remember:** Users can type names instead of numerical IP addresses.
3. **Helps locate servers:** DNS helps devices find the server hosting a website or service.
4. **Supports Internet communication:** It is an essential part of accessing websites and online services.
5. **Reduces the need to remember IP addresses:** Users do not need to remember addresses such as 142.250.x.x.
6. **Uses DNS servers:** DNS queries are handled by specialized DNS servers.
7. **Provides name resolution:** The process of finding an IP address from a domain name is called **name resolution**.
8. **Supports different record types:** DNS stores records such as **A, AAAA, MX, and CNAME**.
9. **Supports email delivery:** MX records help identify the mail servers responsible for a domain.
10. **Improves network usability:** DNS makes Internet services easier and more convenient to access.

How DNS Works

When a user enters a website name:

Domain Name → DNS Query → DNS Server → IP Address → Website Server

For example:

www.example.com → DNS → 93.184.216.34

The user's device can then use the IP address to communicate with the appropriate server.

Advantages of DNS

- Easy to remember domain names.
- Allows websites to change server IP addresses without changing their domain names.
- Supports efficient Internet communication.
- Helps locate web and mail servers.
- Provides a hierarchical and distributed naming system.

Conclusion

DNS is the system that translates domain names into IP addresses. It acts like a **telephone directory for the Internet**, allowing users to access websites and services using easy-to-remember names instead of numerical IP addresses.

12. Why is Networking Knowledge Important for Cybersecurity?

Definition

Networking knowledge is the understanding of how computers, devices, servers, and networks communicate with each other. It is very important in **cybersecurity** because most cyberattacks happen through networks.

Importance of Networking Knowledge in Cybersecurity

1. **Understanding Network Communication**

   Networking knowledge helps security professionals understand how data travels between computers, servers, and other devices.

2. **Identifying Network Attacks**

   It helps in understanding attacks such as **DDoS, Man-in-the-Middle (MITM), packet sniffing, and spoofing**.

3. **Using IP Addresses**

   Cybersecurity professionals need to understand **IP addresses** to identify devices, investigate suspicious connections, and trace network activity.

4. **Understanding Ports and Protocols**

   Knowledge of protocols such as **TCP, UDP, HTTP, HTTPS, DNS, and FTP** helps identify normal and suspicious network traffic.

5. **Firewall Configuration**

   Networking knowledge is required to configure firewalls that control incoming and outgoing network traffic.

6. **Network Monitoring**

   Security teams monitor network traffic to detect unusual activities, unauthorized connections, and possible attacks.

7. **Intrusion Detection and Prevention**

   Understanding networking helps professionals use **IDS (Intrusion Detection Systems)** and **IPS (Intrusion Prevention Systems)** to detect and prevent attacks.

8. **Troubleshooting Security Problems**

   When a security issue occurs, networking knowledge helps determine whether the problem is caused by a device, connection, protocol, or network configuration.

9. **Network Security Design**

   It helps in designing secure networks using **VPNs, firewalls, segmentation, access controls, and secure protocols**.

10. **Incident Response**

    During a cyberattack, security professionals use networking concepts to identify the source of the attack, affected systems, and network connections.

Example

Suppose an attacker tries to access a company's server through an unusual port. A cybersecurity professional with networking knowledge can examine the **IP address, port number, protocol, and network traffic** to identify and block the suspicious connection.

Conclusion

Networking knowledge is a **fundamental skill in cybersecurity**. It helps cybersecurity professionals understand network communication, detect attacks, configure security devices, monitor traffic, and respond to security incidents effectively. Therefore, a strong understanding of networking is essential for a successful career in cybersecurity.

13. Write your computer's private IPv4 address.

How to find it in Windows:

1. Press **Windows + R**.
2. Type cmd and press **Enter**.
3. Type:

ipconfig

4. Look for **IPv4 Address** under your active Wi-Fi or Ethernet adapter.

Example Private IPv4 Address: 192.168.1.10

IP ADDRESS: 10.150.34.91

14. Write your Default Gateway.

How to find it in Windows:

1. Press **Windows + R**.
2. Type cmd and press **Enter**.
3. Type:

ipconfig

4. Find **Default Gateway** under your active Wi-Fi or Ethernet adapter.

Example Default Gateway: 192.168.1.1

DEFAULT GATEWAY ADDRESS : 10.150.34.120

15. Write your DNS Server.

How to find it in Windows:

1. Press **Windows + R**.
2. Type cmd and press **Enter**.
3. Type:

ipconfig /all

4. Find **DNS Servers** under your active Wi-Fi or Ethernet adapter.

Example DNS Server: 192.168.1.1

DNS Server : 10.150.34.120

```
C:\Users\acer>tracert 10.150.[redacted]

Tracing route to PM [10.150.[redacted]]
over a maximum of 30 hops:

  1    <1 ms    <1 ms    <1 ms  PM [10.150.[redacted]]

Trace complete.

C:\Users\acer>
```

```
C:\Users\acer>ipconfig

Windows IP Configuration


Wireless LAN adapter Local Area Connection* 8:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 9:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . :
   Temporary IPv6 Address. . . . . . : [redacted]:40f4:[redacted]:23cc:[redacted]:e58a:[redacted]
   Link-local IPv6 Address . . . . . : fe80::88[redacted]0:a567:5735
   IPv4 Address. . . . . . . . . . . : 10.150.[redacted]
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : [redacted]
                                       10.150.[redacted]

C:\Users\acer>
```

```
C:\Users\acer>ping 10.150.[redacted]

Pinging 10.150.[redacted] with 32 bytes of data:
Reply from 10.150.[redacted]: bytes=32 time<1ms TTL=128
Reply from 10.150.[redacted]: bytes=32 time<1ms TTL=128
Reply from 10.150.[redacted]: bytes=32 time<1ms TTL=128
Reply from 10.150.[redacted]: bytes=32 time<1ms TTL=128

Ping statistics for 10.150.[redacted]:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```